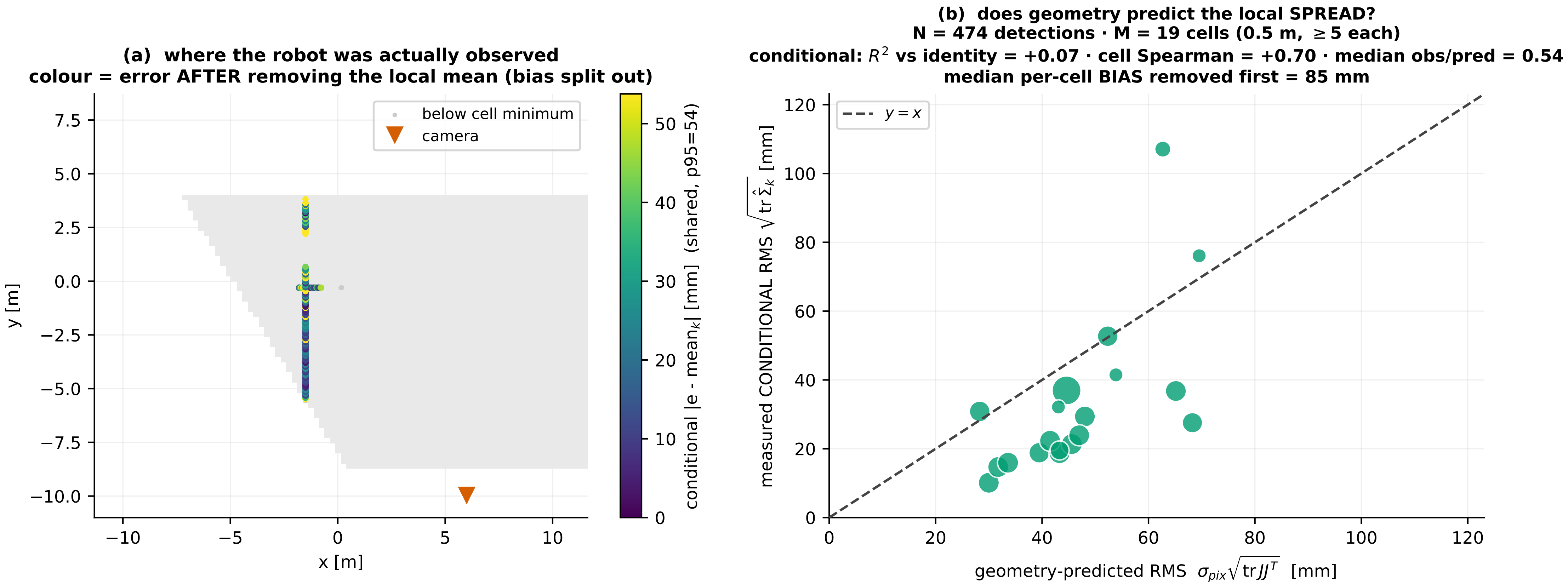
Camera C — deployed-corrected residual



Point area in (b) scales with detections in the cell. sigma_pix is fitted on leave-region-out folds, so each cell's prediction is out-of-region. Ground truth positions the dots in (a) and measures the residual; it never enters a projection, Jacobian or covariance. READ THE LIMIT BEFORE THE RESULT: the sampled ribbon in (a) never covers the footprint in two dimensions, so range, image row and projection conditioning are collinear along it. A weak or negative R^2 here therefore does NOT show that geometry fails -- it shows this dataset cannot test it. The honest conclusion is that straight-route logs are insufficient to decide whether full 2-D projection geometry beats a simpler range effect; that needs a commissioning capture with 2-D coverage and multiple headings per location.